tabreed
India

The Cooling Challenge:

Accelerating District Cooling System (DCS) Adoption In India

(in partnership with APUEA)

India Smart Utility Week, New Delhi, 12 March 2026

tabreed.ae

Tabreed: One of the world's largest cooling utilities

tabreed India

99 plants
In 6 countries

1.57 million RT
of cooling capacity delivered

2.6 billion kWh
Annual energy saved

1.5 million tons
Annual prevention of CO2 emissions

Primary Shareholders

MUBADALA

Government of Abu Dhabi's investment fund

ENGIE

Largest independent energy producer (c. 100 GW) and leader in low-carbon services.

Asia Presence through a dedicated Investment Platform in Singapore, and the required HoldCo and Project SPV structure in India

Project design and delivery

- Technology and OEM agnostic
- Strong R&D and Innovation Focus
- Life-cycle cost view

Financing & capital structure

- Listed on Dubai Fin. Market (DFM)
- **$4 BN** Consolidated Asset Base
- Investment Grade (Fitch-BBB)
- Green Bonds Financing Framework
- Long term capital recovery tariffs

Operations track record

- SLA/KPI based service delivery
- Automation, Unmanned, Central teams
- 27 years, oldest plant in operation
- 99.9% cumulative average reliability

Centralised maintenance

- In-house with minimal OEM reliance
- Reliability Centred Processes

Exclusive provider of district cooling services to several iconic developments

Burj Khalifa & Downtown Dubai

Sheikh Zayed Grand Mosque

Yas Marina Circuit

Yas Mall

Cleveland Clinic Abu Dhabi

Abu Dhabi Global Market Square

Dubai Metro

Ferrari World

SeaWorld Abu Dhabi

Dubai Parks and Resorts

Yas Waterworld

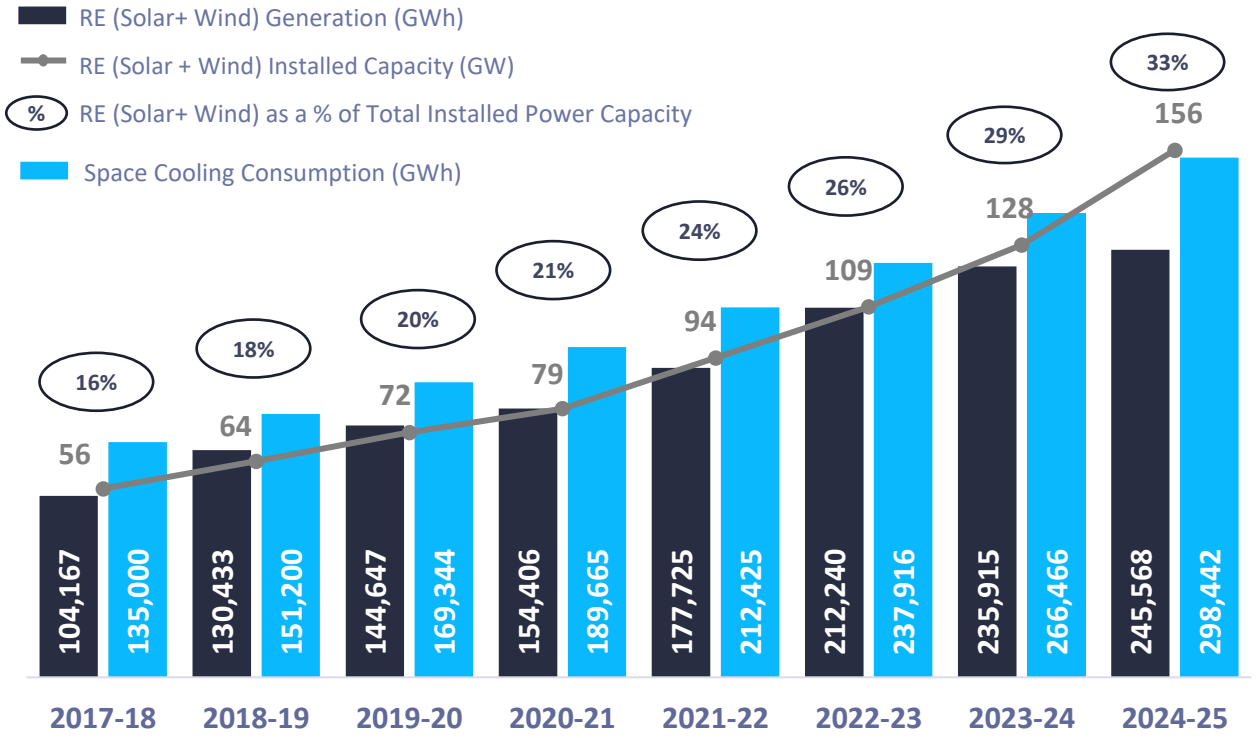
Louvre Abu Dhabi

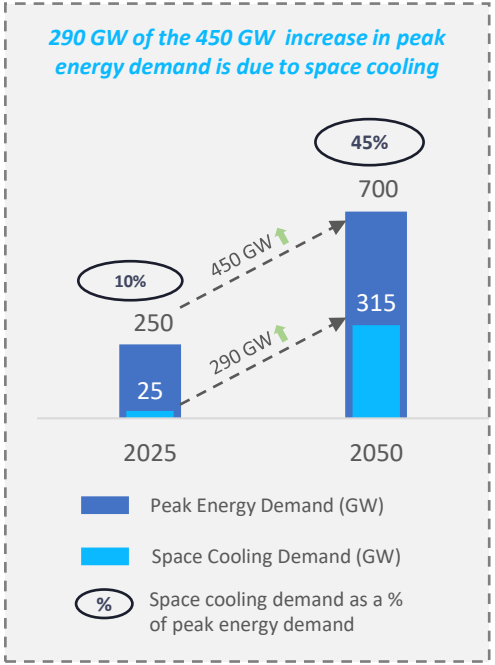
Jabal Omar

Amwaj

Bahrain Financial Harbor

By 2050, 45% of India's peak power demand will come from cooling





Source: CEA, ICAP, IEA, IRENA

District Cooling is already happening in India

Whilst there are more and more precedents coming up for cooling load aggregation in India, what can really push the energy efficiency boundaries would be the **merchant-model DCS projects**

Ownership models		Captive Model	Merchant Model
Single	Public	✓ University campuses, AAI owned airports	✓ GIFT City
Single	Private	✓ 150+ campus captive DCS plants *(5-15kRT)*	✗
Hybrid	PPP	✓ All PPP owned major airports.	✗
Hybrid	Tender- based Concessions	✓ DCS for Amaravati Govt Complex	✓ DCS for Hyderabad Pharma City

150+ campus captive DCS plants (5-15kRT)

Project Design & Management

- **Growing market eco-system** for green building design and execution.
- **In-house teams** for design coordination, project, procurement, and supply chain mgmt.
- **Wide spectrum** of technology choices.

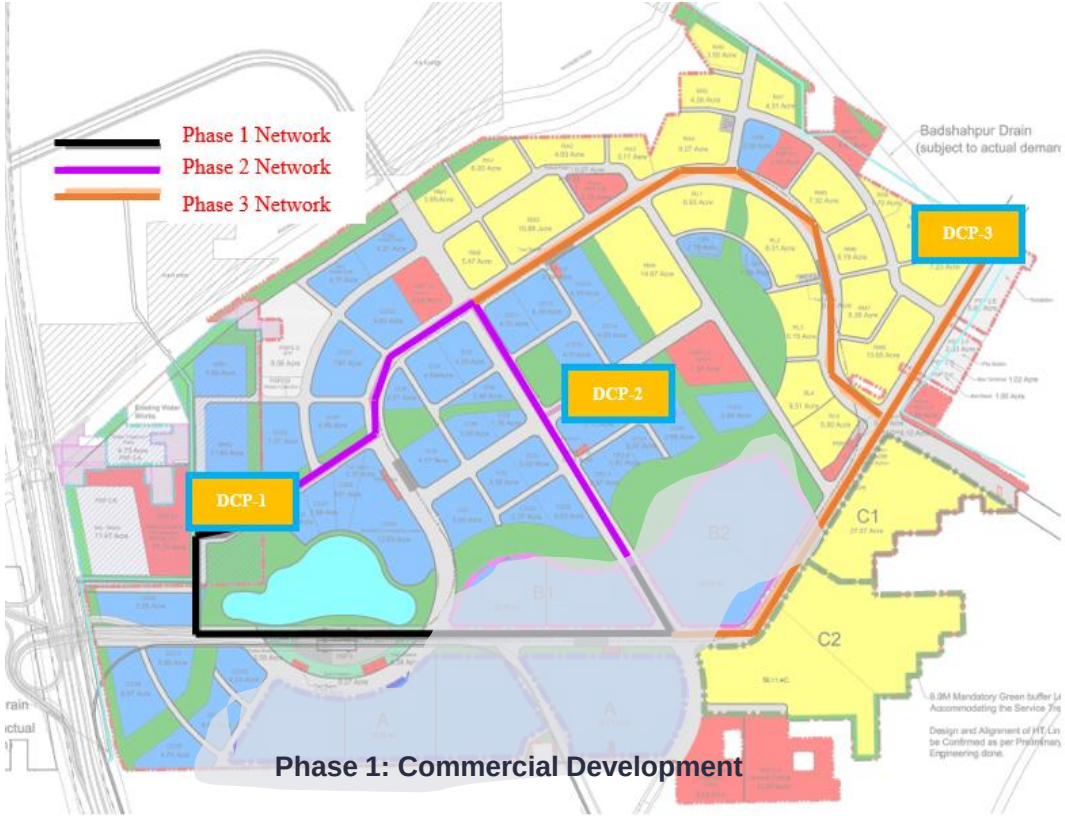
Operations & Maintenance

- Integrated **Facility Management** with increasing market capture by global independent property management companies.
- **Integrated technical services** extend beyond cooling (water, sewage, power, back-up, etc.)
- Large **city specific eco-system** of service providers.
- Heavy **dependence on OEMs** for maintenance.

a) Case Studies – Govt. led masterplans

HSIIDC’s Global City Gurugram

Mixed use masterplan comprising of residential and commercial zones,. Based on anticipated development mix, likely demand for cooling at Global City is estimated at **163,270 RT** for 93.1 mn sft.



Phase 1: Commercial Development

60,000 RT (40%) reduction in cooling demand

110 MW (60%) reduction in power demand

100 million kWh (25%) reduction in annual power consumption

90,000 tons annually (~25%) of carbon emissions

> USD 150 Mn in DC investments

Mega Re-development Projects in NCR

Redevelopment of Sarojini Nagar

288 Acres; Residential & commercial spaces to be developed

Cooling: 10-15,000 TR

Redevelopment of East Kidwai Nagar

86 acres; 1.3 mn SFT of commercial office space and residential units

Cooling: 5,000 TR

Redevelopment of Nauroji Nagar

6 mn SFT of commercial area

Cooling: 17-20,000 TR

Some key projects under development include:

- Redevelopment of GPRA Colony at Srinivaspuri
- Ladakh Bhawan (Leh Wing) At Dwarka In New Delhi
- Construction of Atal Akshay Urja Bhawan CGO Complex, Lodhi Road, New Delhi
- India International Convention and Expo Centre
- New Campus development for Ambedkar University
- Construction of CISF Residential Quarters at Saket
- Nangloi Group Housing Residential Project at Roshanara Road

TIDCO FinTech City, Chennai – Public-Private Partnership for DCS

Possible Location & Network for DCS at FinTech City

6,000 RT (35%) reduction in cooling demand

10-15 MW (60%) reduction in power demand

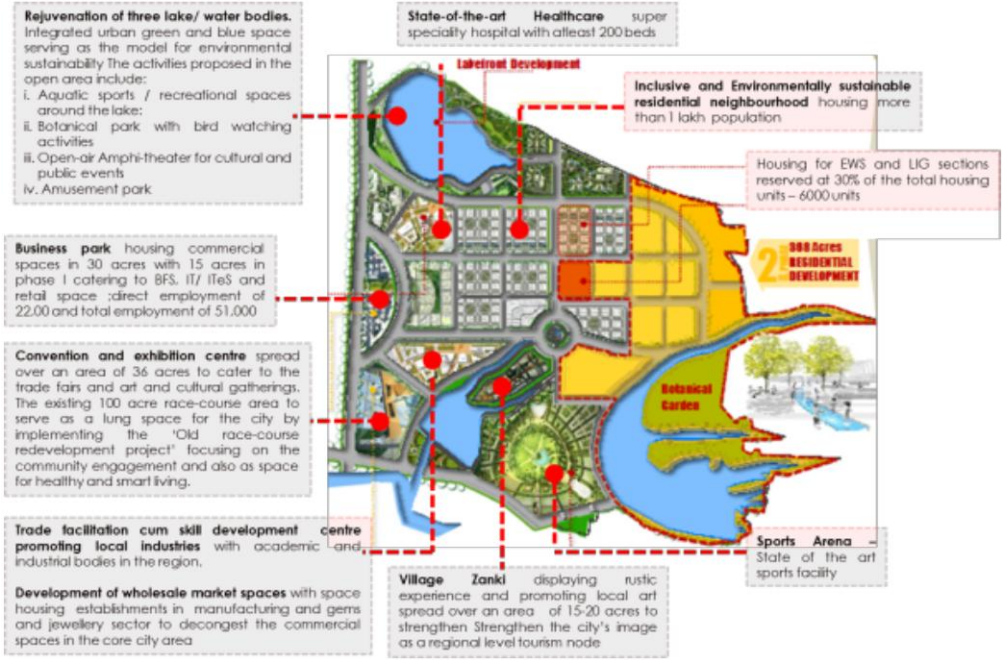
16 million kWh (30%) reduction in annual power consumption

15,000 tons annually (~30%) of carbon emissions

> USD 20 Mn in DC investments

TIDCO floated a request for proposal to conduct a PFR for incorporating DCS in Fin-Tech City masterplan. However, due to delays in receiving suitable bids, DCS was dropped in Phase 1.

Rajkot Smart City incorporated DCS in masterplan but on-hold since then



Area based components to be connected to DCS for cooling: Business Park, Convention & exhibition centre, wholesale market place, hospital, sports arena, etc.

- Rajkot included district cooling in their planning of ABD.
- The new smart city area will require all non-residential buildings and HIG residential buildings to connect to the DCS.

Estimated DCS Details

USD 40 MN
Potential investment capacity for DCS

32,000 RT
Peak cooling demand proposed to be served from a District Cooling plant

Cost & Energy Efficiency Impact

21 MW
Reduction of power demand

28 mn kWh
annual reduction in power consumption at steady state

25% reduction
in lifecycle cooling costs

Current Status

Change in administration and no direct mandate for administration appear to be reasons for no progress.

Central Vista Redevelopment – Missed Opportunity

Redefining Marquee Landscape - Investment potential of INR 250 crores for Central Vista redevelopment

Existing Parliament: Structural enhancements
Built-up area: 45,000 sq.mt

Existing Secretariat: Converted to museums

New Office Buildings for Ministries

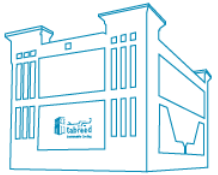
+

New state-of-the-art Parliament building;
Built-up area: 65,000 sq.mt

New Secretariat: World-class high rise office building

Residences for Dignitaries

=



DCS Phase 1: **2,600-3,000** RT

Future DCS Phases: 10,000-12,000 RT
Expansion in modular manner

Kochi: Infopark and Smart City

Kochi InfoPark Phase 1 – 100 acres, 7 mn SFT BUA (Fully developed)

Kochi InfoPark Phase 2 – 160 acres, 8 mn SFT BUA (completion expected in 2030)

Smart City Kochi by Prestige– 246 acres, 7.4 mn SFT BUA (Ph2 under construction)

Replace existing cooling infrastructure with sustainable district cooling plants

30,000 RT
Existing cooling demand (estimated)

10-15 MW
Power demand reduction

22 mn kWh
Annual reduction in energy consumption

20,000 RT
Proposed (Installed) District Cooling plant capacity

USD 35 Mn
DCS Investment Opportunity

Single DC plant can provide cooling to both the adjacent development projects, with significant diversity benefits, given Kochi Smart City is a mixed-use development

60,000 RT
Total cooling demand

30 MW
Power demand reduction

45 mn kWh
Annual reduction in energy consumption

35,000 RT (+ 5000 RT TES)
Proposed (Installed) District Cooling plant capacity

USD 70 Mn
DCS Investment Opportunity

b) Case Studies – Retrofit Opportunities in High-Density Areas

Bandra Kurla Complex (BKC), Mumbai: Home to prominent buildings

- ✓ ***30+ towers*** *with up to* ***50% certified green***
- ✓ ***Mixed-use developments*** *with* ***BUA of 20-30 Mn sft.***
- ✓ ***Average age*** *of buildings* ***15-20 years***

- **Current approach:** Basement cooling plants (air & water), operated by IFMs, no efficiency incentive (cost plus models – charged to end-users)
- **DCS/CAAS Approach:** Optimize, retrofit, aggregate, lifecycle approach for higher efficiencies

DCP 1

14,500 RT	2,500 RT
Mechanical Capacity	TES Tank

DCP 2

29,500 RT	2,500 RT
Mechanical Capacity	TES Tank

28,000 RT (40%) reduction in installed mechanical capacity

30 MW (45%-50%) reduction in Peak Energy Demand

30 million kWh (18%) reduction in Annual Energy Consumption

25,000 tons annual elimination of CO2 emissions

USD 85 MN in DC investments

Campus to Merchant DCS (Retrofit Opportunity) – Taramani, Chennai

Development (IT Parks)	Gross Leasable Area (MN SFT)	Cooling Requirement (RT)
Tata Ramanujan **(Anchor)**	4.6	12,100
Ascendas IT Park	2.1	2,250
Brigade WTC	1.8	2,000
DLF Downtown Taramani	1.9	6,000
Global Infocity Park	2.6	3,300
RMZ Millenia	2.5	15,800
Tidel Park	1.3	10,000
Total	**17.0**	**51,500**

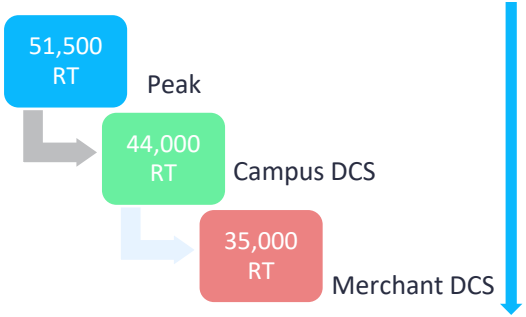
Regulatory/Policy Enablers

- ✓ Policy/guidelines for routes/easement for networks (Urban Planning)
- ✓ Policy for direct power procurement by DCS utility (Ministry of Power)
- ✓ Applicability of GST for provision of chilled water (Ministry of Finance)
- ✓ Mandating for District Cooling

Campus to Merchant DCS Gains

- **16,500 RT:** Installed Capacity reduction
- **25 MW:** Peak power demand reduction
- **38 Mn kWh:** Annual energy consumption savings
- **35,000 (25%)** lower carbon emissions annually
- **USD 60 MN** in DC investments

Estimated Cooling Capacities



Barriers to address for Merchant Model for DCS

Long-Term Market Development Efforts for Merchant DCS

A Identify opportunities

- Scope and assess:
  - Urban greenfield masterplans,
  - Infrastructure development corridors
  - Urban retrofit projects

B Secure initial interest

- Prospective customers:
  - SIDCs (TGIIC, HSIIDC, TIDCO), ULBs (APCRDA)
  - Govt. entities (NBCC, CPWD, Petronet)

C Technical support to establish viability

- Master planning review
- Infrastructure Review
- Establish feasibility
- Circularity and resource use and planning

D Commercial & Contractual Strategy

- Evaluate DC opportunity in terms of size, capex requirement
- Contract structuring /framework support

E DCS Concession Tender, Award & Construction

- Pre-bid queries/addendum to tenders
- Complexities with govt change, etc.

Tabreed's Experience in India to date

Government bodies & current regulatory framework:

- Limited appetite to consider new market models
- CPWD & NBCC execute projects on behalf of govt orgs and ministries, who pay for capex; DCS as a utility would entail PPP structures where capex would be funded by cooling provider. No framework in place to govern this process.
- State govt often seeks clarity/guidelines from the centre.
- **Regulatory framework as a deterrent: Power tariff structure, AERA for airports, un-even GST applicability, Easement/Right of Way, Utility plots unavailable**

Developers:

- Resistance to outsourced models; Limited incentive for pre-investment in capacity and efficiency; Residential build-to-sell;
- Lack of standard market metrics to measure efficiency
- No attunement to lifecycle cost; focus on capex opex reduction

Market ecosystem:

- Competent consultants, contractors and workforce
- Local manufacturing of DCS equipment

Lack of minimum bulk offtake at a certain scale and with credit-worthy counterparty:

- **No mechanism for aggregation of demand** from multiple buildings into a single contractual framework with a nodal agency ensuring a **certain minimum offtake** with a **certain minimum creditworthiness**/payment security mechanism

Thank You